THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 3: **World of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of $0.\overline{2} + 0.\overline{5}$ is: 1

(a) $\frac{7}{10}$ **(b)** $\frac{7}{9}$ **(c)** $\frac{8}{9}$ **(d)** $\frac{2}{3}$

2. $0.\overline{13} + 0.\overline{31}$ equals: 1

(a) $\frac{1}{3}$ **(b)** $\frac{4}{9}$ **(c)** $\frac{2}{5}$ **(d)** $\frac{5}{9}$

3. The smallest natural number n for which $\frac{n}{120}$ has a terminating decimal expansion is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 5

4. If $x = 2 + \sqrt{3}$ with $2^2 - 3 = 1$, then $x + \frac{1}{x} =$ 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

5. $1.\overline{27}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{13}{11}$ **(b)** $\frac{15}{11}$ **(c)** $\frac{14}{11}$ **(d)** $\frac{7}{6}$

6. The value of $0.\overline{9}$ is: 1

(a) 1 **(b)** just less than 1

(c) 0.99 **(d)** 0.9

7. The length of the repeating block of $\frac{1}{41}$ is: 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 4

8. $0.\overline{18} \times 11$ equals: 1

(a) 2 **(b)** 3 **(c)** $\frac{2}{3}$ **(d)** 1

9. **Assertion (A):** The number 0.375 is a rational number. 1

Reason (R): Every terminating decimal can be written in the form $\frac{p}{q}$ with integers p, q and $q \neq 0$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The number $\sqrt{2}$ is irrational. 1

Reason (R): The decimal expansion of an irrational number is non-terminating and non-repeating.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Express $0.\overline{17} + 0.\overline{71}$ as a single fraction in lowest terms. 2

12. Find the smallest natural number n such that $\frac{n}{420}$ has a terminating decimal expansion. 2

13. If $x = 6 + \sqrt{35}$, find $\frac{1}{x}$ and $x + \frac{1}{x}$. 2

14. Express $0.\overline{29} + 0.\overline{92}$ as a single fraction in lowest terms. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Express $0.4\overline{7}$ as a fraction in lowest terms and verify by long division. 3

16. If $x = 2 + \sqrt{3}$, find $x^2 + \frac{1}{x^2}$. 3

17. Given that $\sqrt{2}$ is irrational, prove that $3 + \sqrt{2}$ is irrational. 3

OR

Express $5.3\overline{2}$ as a fraction in lowest terms and verify by long division. 3

18. If $x = 3 + \sqrt{8}$, find $x^2 + \frac{1}{x^2}$. 3

OR

Given that $\sqrt{3}$ is irrational, prove that $5 + \sqrt{3}$ is irrational. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Mrs Gupta writes the fractions $\frac{7}{16}$, $\frac{5}{12}$ and $\frac{11}{25}$ on the board and asks the class to find their decimal expansions. 5

(i) Write $\frac{7}{16}$ as a decimal.

(ii) Write $\frac{11}{25}$ as a decimal.

(iii) Express the denominator 12 in prime factors.

(iv) Which of the three fractions have terminating decimal expansions? Give the reason.

20. Find the decimal expansion of $\frac{5}{13}$ by long division, state the repeating block and its length, and hence write $\frac{5}{13}$ as a repeating decimal. 5

OR

Simplify $(\sqrt{11} + \sqrt{3})^2 + (\sqrt{11} - \sqrt{3})^2$ and $(\sqrt{11} + \sqrt{3})(\sqrt{11} - \sqrt{3})$. Which are rational? 5

END OF QUESTION PAPER — **Total: 40 Marks**